Titanic Dataset: Detailed Analysis Report

1. Introduction

This report contains a comprehensive analysis of the Titanic dataset using Python and popular data science libraries like Pandas, NumPy, Matplotlib, and Seaborn. The objective is to explore the dataset, uncover meaningful insights, and understand the factors influencing passenger survival.

2. Code Section

```
>>> Code:
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Code Section

```
>>> Code:
# Load the dataset

path = ("C:/Users/satish/Downloads/intern_task-5/train.csv")

df = pd.read_csv(path)

print(df)

>>> Output:
PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                 Name     Sex   Age  SibSp  \
0                             Braund, Mr. Owen Harris    male  22.0      1
1    Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                              Heikkinen, Miss. Laina  female  26.0      0
3         Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0
..                                               ...     ...   ...    ...
886                            Montvila, Rev. Juozas    male  27.0      0
```

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```
887                      Graham, Miss. Margaret Edith  female  19.0      0
888           Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                              Behr, Mr. Karl Howell    male  26.0      0
890                                Dooley, Mr. Patrick    male  32.0      0

     Parch            Ticket     Fare Cabin Embarked
0        0         A/5 21171   7.2500   NaN        S
1        0          PC 17599  71.2833   C85        C
2        0  STON/O2. 3101282   7.9250   NaN        S
3        0            113803  53.1000  C123        S
4        0            373450   8.0500   NaN        S
..     ...               ...      ...   ...      ...
886      0            211536  13.0000   NaN        S
887      0            112053  30.0000   B42        S
888      2        W./C. 6607  23.4500   NaN        S
889      0            111369  30.0000  C148        C
890      0            370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

4. Code Section

```
>>> Code:
# Basic Exploration of dataset

print("\nData Information:")
print(df.info())
print("*"*100)

print("\nData Description:")
print(df.describe())
print("*"*100)

print("\nValue Counts For 'Survived':")
print(df["Survived"].value_counts())
print("Here '0' refers Died")
print("Here '1' refers Alive")

>>> Output:
Data Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
```

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```
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
**********************************************************************************************
**

Data Description:
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
**********************************************************************************************
**

Value Counts For 'Survived':
Survived
0    549
1    342
Name: count, dtype: int64
Here '0' refers Died
Here '1' refers Alive
```

5. Code Section

```
>>> Code:
# Checking the Missing Values
```

```
print("\nMissing Values Before Replacement")
print(df.isna().sum())

>>> Output:
Missing Values Before Replacement
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

6. Code Section

```
>>> Code:
# Here Age, Cabin and Embarked has missing values
```

7. Code Section

```
>>> Code:
# Replace Missing Values
# For Age replace with median

df["Age"] = df["Age"].fillna(df["Age"].median())
```

8. Code Section

```
>>> Code:
# For Embraked replace with mode

df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

9. Code Section

```
>>> Code:
# For Cabin replace with

df["Cabin"] = df["Cabin"].fillna("Unknown")
```

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10. Code Section

```
>>> Code:
# Checking missing values for filling the values

df.isna().sum()

>>> Output:
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

11. Code Section

```
>>> Code:
df.dtypes

>>> Output:
PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

12. Code Section

```
>>> Code:
# Data Visualization using Matplotlib and seaborn

# Histogram
```

```
df.hist(figsize = (12, 10), color = 'skyblue', edgecolor = 'blue')
plt.suptitle('Histograms of Titanic Features', fontsize = 16)
plt.show()

>>> Output:
[Visual Output]
```

13. Code Section

```
>>> Code:
# Observation 1: PassengerId distribution
print("PassengerId is uniformly distributed.")
print("*"*120)

# Observation 2: Survived distribution
survival_counts = df['Survived'].value_counts()
print("\nSurvival Counts:\n", survival_counts)
if survival_counts[0] > survival_counts[1]:
    print("More passengers did NOT survive.")
else:
    print("More passengers survived.")
print("*"*120)

# Observation 3: Pclass distribution
pclass_counts = df['Pclass'].value_counts()
print("\nPassenger Class Counts:\n", pclass_counts)
print(f"Most passengers belong to class {pclass_counts.idxmax()}.")
print("*"*120)

# Observation 4: Age distribution
print("\nAge Statistics:")
print(df['Age'].describe())
print("Most passengers are between 20 and 40 years old.")
print("*"*120)

# Observation 5: SibSp distribution
sibsp_counts = df['SibSp'].value_counts()
print("\nSibSp (Siblings/Spouses aboard) Counts:\n", sibsp_counts)
print("Most passengers had no siblings or spouses aboard.")
print("*"*120)

# Observation 6: Parch distribution
parch_counts = df['Parch'].value_counts()
print("\nParch (Parents/Children aboard) Counts:\n", parch_counts)
print("Most passengers had no parents or children aboard.")
print("*"*120)

# Observation 3: Fare distribution
```

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```
print("Fare distribution is right-skewed, with most fares under $100 and a few outliers with very high fares.")

>>> Output:
PassengerId is uniformly distributed.
**************************************************************************************************
**********************

Survival Counts:
 Survived
0    549
1    342
Name: count, dtype: int64
More passengers did NOT survive.
**************************************************************************************************
**********************

Passenger Class Counts:
 Pclass
3    491
1    216
2    184
Name: count, dtype: int64
Most passengers belong to class 3.
**************************************************************************************************
**********************

Age Statistics:
count    891.000000
mean      29.361582
std       13.019697
min        0.420000
25%       22.000000
50%       28.000000
75%       35.000000
max       80.000000
Name: Age, dtype: float64
Most passengers are between 20 and 40 years old.
**************************************************************************************************
**********************

SibSp (Siblings/Spouses aboard) Counts:
 SibSp
0    608
1    209
2     28
4     18
3     16
8      7
5      5
```

```
Name: count, dtype: int64
Most passengers had no siblings or spouses aboard.
********************************************************************************************************************
**********************

Parch (Parents/Children aboard) Counts:
 Parch
0    678
1    118
2     80
5      5
3      5
4      4
6      1
Name: count, dtype: int64
Most passengers had no parents or children aboard.
********************************************************************************************************************
**********************
Fare distribution is right-skewed, with most fares under $100 and a few outliers with very high fares.
```

14. Code Section

```
>>> Code:
# Boxplots

plt.figure(figsize = (8,6))
sns.boxplot(x = 'Survived', y = 'Age', data = df)
plt.title("Boxplot of Age by 'Survived'")
plt.show()
print("Here '0' refers Died")
print("Here '1' refers Alive")

>>> Output:
[Visual Output]Here '0' refers Died
Here '1' refers Alive
```

15. Code Section

```
>>> Code:
# Observations
died_age_stats = df[df['Survived'] == 0]['Age'].describe()
survived_age_stats = df[df['Survived'] == 1]['Age'].describe()

print("Age Statistics for passengers who died (Survived=0):")
print(died_age_stats)
print("*"*120)

print("\nAge Statistics for passengers who survived (Survived=1):")
```

```
print(survived_age_stats)
print("*"*120)

# Example insights:
if died_age_stats['mean'] > survived_age_stats['mean']:
    print("\nObservation: On average, passengers who died were older than those who survived.")
else:
    print("\nObservation: On average, passengers who survived were older than those who died.")

if died_age_stats['25%'] > survived_age_stats['25%']:
    print("Younger passengers were more likely to survive.")
else:
    print("Survivors include many young passengers.")

>>> Output:
Age Statistics for passengers who died (Survived=0):
count    549.000000
mean      30.028233
std       12.499986
min        1.000000
25%       23.000000
50%       28.000000
75%       35.000000
max       74.000000
Name: Age, dtype: float64
************************************************************************************************************
**********************

Age Statistics for passengers who survived (Survived=1):
count    342.000000
mean      28.291433
std       13.764425
min        0.420000
25%       21.000000
50%       28.000000
75%       35.000000
max       80.000000
Name: Age, dtype: float64
************************************************************************************************************
**********************

Observation: On average, passengers who died were older than those who survived.
Younger passengers were more likely to survive.
```

16. Code Section

```
>>> Code:
# ScatterPlot
```

```
plt.figure(figsize = (8,6))
sns.scatterplot(x = 'Age', y = 'Fare', hue = 'Survived', data = df)
plt.title("Scatterplot of Age 'V/s' Fare by Survival")
plt.show()
print("Here '0' refers Died")
print("Here '1' refers Alive")

>>> Output:
[Visual Output]Here '0' refers Died
Here '1' refers Alive
```

17. Code Section

```
>>> Code:
# Observations
print("Observations:")
print("1. Majority of passengers paid fare below 100.")
print("2. Younger passengers (age < 15) have a higher survival rate (more orange dots).")
print("3. High fare passengers (fare > 200) show more survival cases (orange).")
print("4. Elderly passengers (age > 60) have mixed survival outcomes.")
print("5. Survival seems more frequent in passengers with moderate age and higher fare.")

# Optional: You can break this down into more analytical code if needed.

>>> Output:
Observations:
1. Majority of passengers paid fare below 100.
2. Younger passengers (age < 15) have a higher survival rate (more orange dots).
3. High fare passengers (fare > 200) show more survival cases (orange).
4. Elderly passengers (age > 60) have mixed survival outcomes.
5. Survival seems more frequent in passengers with moderate age and higher fare.
```

18. Code Section

```
>>> Code:
# Pairplot

sns.pairplot(df[['Survived', 'Age', 'Fare', 'Pclass']], hue = 'Survived')
plt.suptitle("Pairplot of Selected Features", y =  1.02)
plt.show()

>>> Output:
[Visual Output]
```

19. Code Section

```
>>> Code:
# Observations based on the pairplot
```

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```
observations = """
1. Fare vs Pclass: Passengers in lower classes (e.g., Pclass 3) generally paid lower fares, while higher class passengers (Pclass 1) paid more.
2. Fare vs Survived: Survived passengers tend to have paid higher fares, indicating a potential survival advantage for wealthier passengers.
3. Age Distribution: There is no clear age boundary for survival, but many survivors are observed in younger age ranges.
4. Pclass vs Survived: Higher survival rate is seen in Pclass 1 compared to Pclass 3, suggesting class significantly impacted survival.
5. Age vs Fare: No strong correlation, but most passengers who paid high fares were also from a smaller age group (middle-aged).
"""

print("Observations from the pairplot:")
print(observations)

>>> Output:
Observations from the pairplot:

1. Fare vs Pclass: Passengers in lower classes (e.g., Pclass 3) generally paid lower fares, while higher class passengers (Pclass 1) paid more.
2. Fare vs Survived: Survived passengers tend to have paid higher fares, indicating a potential survival advantage for wealthier passengers.
3. Age Distribution: There is no clear age boundary for survival, but many survivors are observed in younger age ranges.
4. Pclass vs Survived: Higher survival rate is seen in Pclass 1 compared to Pclass 3, suggesting class significantly impacted survival.
5. Age vs Fare: No strong correlation, but most passengers who paid high fares were also from a smaller age group (middle-aged).
```

20. Code Section

```
>>> Code:
# Heatmap (Correlation)
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes(include=['float64', 'int64']).corr(), annot=True, cmap='coolwarm', linewidths=0.5)
plt.title("Heatmap of Correlation")
plt.show()

>>> Output:
[Visual Output]
```

21. Code Section

```
>>> Code:
#Observations
print("\nKey Observations:")
# Top positively correlated with 'Survived'
```

```
print("1. Survived is moderately positively correlated with Fare (0.26).")
print("2. Survived has weak positive correlation with Parch (0.082).")
# Top negatively correlated with 'Survived'
print("3. Survived is moderately negatively correlated with Pclass (-0.34).")
print("4. Survived has weak negative correlation with Age (-0.065).")
# Other noticeable correlations
print("5. Pclass and Fare have a strong negative correlation (-0.55).")
print("6. SibSp and Parch show a moderate positive correlation (0.41).")

>>> Output:
Key Observations:
1. Survived is moderately positively correlated with Fare (0.26).
2. Survived has weak positive correlation with Parch (0.082).
3. Survived is moderately negatively correlated with Pclass (-0.34).
4. Survived has weak negative correlation with Age (-0.065).
5. Pclass and Fare have a strong negative correlation (-0.55).
6. SibSp and Parch show a moderate positive correlation (0.41).
```

22. Code Section

```
>>> Code:
# Observations (printed directly here)
observations = """
Observations:
- Survival Rate: More people did not survive (0) compared to survivors (1).
- Missing Values: After replacement, no missing values remain.
- Age: Younger passengers had slightly higher survival rates.
- Fare: Higher fare seems associated with higher survival.
- Pclass: Lower class (higher number) had lower survival rates.
- Correlations: Fare and Pclass negatively correlated; Survived positively correlated with Fare.
"""
print(observations)

# Summary
summary = """
Summary of Findings:
- Younger and wealthier passengers had higher survival probabilities.
- First class passengers survived more compared to second and third classes.
- Age and Fare show some positive correlation with survival.
- Missing values have been handled properly.
- Fare is skewed; could consider transformations.
"""
print(summary)

>>> Output:
Observations:
- Survival Rate: More people did not survive (0) compared to survivors (1).
- Missing Values: After replacement, no missing values remain.
- Age: Younger passengers had slightly higher survival rates.
```

- Fare: Higher fare seems associated with higher survival.
- Pclass: Lower class (higher number) had lower survival rates.
- Correlations: Fare and Pclass negatively correlated; Survived positively correlated with Fare.

Summary of Findings:
- Younger and wealthier passengers had higher survival probabilities.
- First class passengers survived more compared to second and third classes.
- Age and Fare show some positive correlation with survival.
- Missing values have been handled properly.
- Fare is skewed; could consider transformations.

23. Code Section

>>> Code: